A Two-Dimensional Area Law from a Global Spectral Gap

Positive Filters, Loss-Reduced Localization, and Physical Error Budgets

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Abstract

The source replaces a globally gapped spin Hamiltonian by positive ground-annihilating constraints before using replica transport and entropy amplification. We refine this replacement at the physical operator-norm level. A normalized autocorrelation filter gives a positive time kernel with unit integral, eliminating a filter-dependent normalization loss. A dimension-independent resolvent modulus for Hermitian absolute values replaces the first square-root localization loss by a logarithmic loss; the later channel square roots are retained. We separate scalar spectral, Hilbert–Schmidt, operator, and completely bounded error norms, and account explicitly for finite-time leakage, total truncation, ground-state motion, and controlled-subsystem entropy. The local estimates do not establish the source's replica limits, volume-independent collar information, geometric entropy cancellation hypotheses, or a new global area law. Classical Fourier positivity, functional calculus, and perturbation tools are credited. Contribution assessment: C3/5, loss-reduced positive-constraint localization and norm-correct approximation budgets sharpen the source's physical filtering interface without independently proving its global entropy theorem.

1 Physical setting and the source claim

Let Λ be a finite induced subgraph of the square lattice, with induced graph distance d_Λ, infinite between different components. The physical Hilbert space is $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$. The source [1] assumes a Hermitian Hamiltonian

$$H = \sum_i h_i, \qquad \operatorname{supp} h_i \subseteq X_i, \quad \operatorname{diam}_{d_\Lambda} X_i \le R, \quad \|h_i\| \le J,$$

with one term per designated nonempty support, fixed integers $q \ge 2, R \ge 1$, and $J > 0$. This bounds anchor multiplicity and interaction norm per site in terms of R, J, even with holes. It assumes a unique unit ground vector Ω, up to phase, and

$$H\Omega = E_0\Omega, \qquad H - E_0 I \ge \Delta(I - P), \quad P = |\Omega\rangle\langle\Omega|, \quad \Delta > 0. \tag{1}$$

No gap is assumed for every restricted Hamiltonian. The claimed conclusion is

$$S_\Omega(A) \le C(q, R, J, \Delta)|\partial_\Lambda A| \quad \text{for every } A \subseteq \Lambda,$$

uniformly over the domain, cut and admissible interactions. This is an imported global source claim, not the conclusion of our local certificates. The trivial cases $q = 1$, empty domain, or range zero

factorize directly. A cut with no crossing edge is a union of components; uniqueness makes its entropy zero.

All norms on observables are operator norms unless another norm is specified. Entropies use natural logarithms and normalized partial states. For an anchor a_i, put $B_l = N_l^\Lambda(a_i)$ and

$$\mathsf{E}_{i,l}(A) = q^{-|\Lambda\setminus B_l|}\,\mathrm{Tr}_{\Lambda\setminus B_l}(A)\otimes I_{\Lambda\setminus B_l}. \tag{2}$$

This is a unital completely positive contraction, not an unnormalized partial trace. It fixes the ball algebra and the scalar identity. The tensor factors keep their physical site positions. Auxiliary systems below are spectators, with identity maps applied to them. There is no quotient gauge; the phase of a ground vector is chosen only when comparing vectors.

We use the source's graph-distance propagation estimate

$$\|\tau_t(A) - \mathsf{E}_{i,l}\tau_t(A)\| \le \|A\|\min\{2, C_{\mathrm{LR}}e^{v|t|-c_{\mathrm{LR}}l}\}, \qquad \tau_t(A) = e^{itH}Ae^{-itH}, \tag{3}$$

for A supported on X_i. Its proof in the source was read in full: overlapping interaction chains control commutators, and telescoping on-site unitary averages yields (3). Its constants are independent of $|\Lambda|$. Ambient distance cannot replace graph distance in the interaction hypothesis.

2 A positive normalized spectral filter

The following is a classical autocorrelation construction, applied to the source's particular constraint normalization. It is not a new Fourier inversion theorem.

Proposition 2.1 (Unit-mass time kernel)**.** *Fix $a > 0$ with $2a < \Delta$ and an integer $p \ge 1$. There is a real even nonzero $h \in C_c^\infty([-a,a])$ such that, writing $\alpha = p/(p+1)$,*

$$\chi(\omega) = \frac{\int h(s)h(s-\omega)\,ds}{\|h\|_2^2}, \qquad \check{h}(t) = \frac{1}{2\pi}\int h(s)e^{-its}\,ds, \tag{4}$$

$$f(t) = \frac{2\pi}{\|h\|_2^2}|\check{h}(t)|^2, \qquad \chi(\omega) = \int f(t)e^{it\omega}\,dt. \tag{5}$$

Here $f \ge 0$, $\int f = 1$, $\chi(0) = 1$, $\operatorname{supp}\chi \subseteq [-2a, 2a]$, and $f(t) \le C_{p,a}e^{-c_{p,a}|t|^\alpha}$. The map $\Phi(A) = \int f(t)\tau_t(A)\,dt$ is a unital completely positive contraction, including on all spectator extensions.

Proof. Take the bump $h(s) = \exp[-(1-(s/a)^2)^{-p}]$ for $|s| < a$, zero outside. For real $u \in (-1,1)$, a complex disk of radius $\eta_p(1-u^2)$ has $\mathrm{Re}(1-z^2)^{-p} \ge c_p(1-u^2)^{-p}$ when η_p is sufficiently small. Cauchy's derivative estimate and maximization of $w^{-m}e^{-c_p w^{-p}}$ imply $\left\|h^{(m)}\right\|_\infty \le C^{m+1}(m!)^{1+1/p}$. Every fixed derivative tends to zero at the endpoints, so extension by zero is smooth. Integrating by parts m times and choosing $m = \lfloor c|t|^{p/(p+1)}\rfloor$, with small fixed $c > 0$, gives $|\check{h}(t)| \le C'e^{-c'|t|^\alpha}$. These constants may depend on p, a; no uniform limit $p \to \infty$ is asserted.

The convolution formula for the correlation in (4) gives (5) with the displayed 2π normalization. Plancherel gives $\int f = 1$. Support and the value at zero follow directly from the correlation. The operator integral converges in norm, since $f \in L^1$. It averages unitary conjugations with a probability density, which proves positivity, complete positivity, unitality and contractivity on every spectator extension. □

Corollary 2.2 (A filter-independent gap normalization)**.** *Choose $a = \Delta/4$, let $e_i = \langle\Omega, h_i\Omega\rangle$, and set*

$$M_i = \Phi(h_i) - e_iI, \qquad k_i = \frac{|M_i|}{2J}, \qquad H_F = \sum_i k_i. \tag{6}$$

Then

$$0 \le k_i \le I, \quad k_i\Omega = 0, \quad H_F \ge \frac{H - E_0 I}{2J} \ge g(I - P), \qquad g = \frac{\Delta}{2J}. \tag{7}$$

The normalization and this gap lower bound have no dependence on the fixed Gevrey order p, although spatial decay constants do.

Proof. In an energy eigenbasis the filter multiplies the (m, n) entry by $\chi(E_m - E_n)$. Every ground-to-excited difference lies outside its support. Its ground entry is unchanged, so $M_i\Omega = 0$. Also $\Phi(H) = H$ and $\sum_i e_i = E_0$, whence $\sum_i M_i = H - E_0 I$. Contractivity gives $\|M_i\| \le 2J$. Hermitian functional calculus gives $|M_i| \ge M_i$ and equality of their kernels; summing proves (7). □

For an individual term one can instead use the certified bound $\|M_i\| \le \max\{\lambda_{\max}(h_i) - e_i, e_i - \lambda_{\min}(h_i)\}$, because a unital positive map preserves scalar order intervals. A common upper bound on these quantities can replace $2J$. Individual rescaling without retaining the corresponding weights would change the summed domination argument.

The source's original bump gives the valid normalization $c_* = \max\{1, J(1 + \|f\|_1)\}$. The autocorrelation construction removes its $\|f\|_1$ dependence and the unnecessary lower cutoff at one. This is a quantitative choice within the same exact annihilation mechanism, not an independent entropy theorem or a claim that the source filter fails.

3 Absolute values without a first square-root loss

The source bounds $\||A| - |B|\|$ by $\sqrt{2M\,\|A - B\|}$ when both Hermitian norms are at most M. That is valid and sufficient for qualitative stretched-exponential locality. A classical logarithmic continuity mechanism, going back to work such as [2], yields a more informative finite budget. We prove the version used here completely.

Theorem 3.1 (Dimension-independent Hermitian modulus)**.** *Let $A = A^*, B = B^*$, $\|A\|, \|B\| \le M$, $M > 0$, and $\epsilon = \|A - B\| \le 2M$. Define $\omega_M(0) = 0$ and*

$$\omega_M(\epsilon) = \min\left\{M, \frac{2\epsilon}{\pi}\left[2 + \log\frac{2M}{\epsilon}\right]\right\} \quad (0 < \epsilon \le 2M). \tag{8}$$

Then $\||A| - |B|\| \le \omega_M(\epsilon)$. The estimate is independent of matrix dimension, including after tensoring both matrices with an identity.

Proof. For a Hermitian matrix,

$$|A| = \frac{2}{\pi}\int_0^\infty A^2(A^2 + s^2 I)^{-1}\, ds.$$

Put $F_s(A) = A^2(A^2 + s^2 I)^{-1}$. The integrand difference has three bounds. First, $0 \le F_s(A), F_s(B) \le I$ gives a norm bound one. Second,

$$F_s(A) = I - s\,\mathrm{Im}(A - isI)^{-1}$$

and the resolvent identity give the bound ϵ/s. Third, using the resolvents of the positive squares gives

$$\|F_s(A) - F_s(B)\| \le \frac{\|A^2 - B^2\|}{s^2} \le \frac{2M\epsilon}{s^2}.$$

The factorization $A^2 - B^2 = A(A-B) + (A-B)B$ does not assume commutation. Integrating the three bounds on $(0, \epsilon)$, $(\epsilon, 2M)$, and $(2M, \infty)$, respectively, gives the second quantity in (8). The first follows from $0 \le |A|, |B| \le MI$. The case $\epsilon = 0$ is immediate. Every estimate used the operator norm and scalar resolvent bounds only. □

The function in (8) is increasing on $[0, 2M]$. Thus a certified upper bound on ϵ, clipped to $2M$, may be substituted. No dimension-independent Lipschitz constant at zero is asserted; the logarithm is retained.

Corollary 3.2 (Constraint and channel budgets)**.** *For M_i in (6), put $M = 2J$ and*

$$a_i(l) = \|M_i - \mathsf{E}_{i,l} M_i\|, \qquad d_i(l) = \min\{1, 2\omega_M(a_i(l))/M\}.$$

Then $k_{i,l} = \mathsf{E}_{i,l} k_i$ is a positive contraction and

$$\|k_i - k_{i,l}\| \le d_i(l). \tag{9}$$

Let $G_i = \sqrt{I - k_i}$, $K_i = \sqrt{k_i}$, and define $G_{i,l}, K_{i,l}$ with $k_{i,l}$. Each root difference is at most $\sqrt{d_i(l)}$. The unital channels $\mathcal{E}_i(X) = G_i X G_i + K_i X K_i$ satisfy

$$\|\mathcal{E}_i - \mathcal{E}_{i,l}\|_{\mathrm{cb}} \le \min\{2, 2\sqrt{2d_i(l)}\}. \tag{10}$$

Proof. Both M_i and its conditional expectation are Hermitian of norm at most M. The operator $|\mathsf{E}_{i,l} M_i|$ lies in the ball algebra and is fixed by $\mathsf{E}_{i,l}$. Contractivity and Theorem 3.1 therefore give

$$\| |M_i| - \mathsf{E}_{i,l}|M_i| \| \le 2 \| |M_i| - |\mathsf{E}_{i,l} M_i| \| \le 2\omega_M(a_i(l)).$$

Divide by M; two positive contractions differ in norm by at most one. For positive operators the standard order proof gives $\left\|U^{1/2} - V^{1/2}\right\| \le \|U - V\|^{1/2}$: use operator monotonicity on $U \le V + \epsilon I$, then reverse the roles. For the channels use the isometries $V_i \xi = G_i \xi \oplus K_i \xi$ and $V_{i,l}$ analogously. Their difference has norm at most $\sqrt{2d_i(l)}$, and $\mathcal{E}_i(X) = V_i^*(X \oplus X)V_i$. Telescoping the two isometries gives (10) on every auxiliary extension. The cap two follows from contractivity. □

If $a_i(l) \le Ce^{-cl^\alpha}$ with fixed constants, then

$$d_i(l) \le C'(1 + l^\alpha)e^{-cl^\alpha}, \qquad \|\mathcal{E}_i - \mathcal{E}_{i,l}\|_{\mathrm{cb}} \le C''(1 + l^\alpha)^{1/2} e^{-cl^\alpha/2}. \tag{11}$$

This retains the input absolute-value decay constant up to a polynomial prefactor. The source's two successive square-root estimates would give a root/channel bound with exponent constant $c/4$ at this comparison stage. We do not compare unspecified global constants, assert an improved entropy exponent, or remove the final square root.

4 Finite-time budgets and exact-annihilation leakage

The time-domain norm is important even in a finite physical system. For integrable kernels $f, \widetilde{f}$, direct integration gives

$$\left\|\Phi_f - \Phi_{\widetilde{f}}\right\|_{\mathrm{cb}} \le \left\|f - \widetilde{f}\right\|_1. \tag{12}$$

This holds independently of the number and spacing of energies. The completely bounded norm here is that of a linear map on observables, not the norm of a scalar function of energy difference.

For the positive probability kernel of Proposition 2.1, let $\tau(T) = \int_{|t|>T} f(t)\,dt$. Equation (3) yields the explicit family of bounds

$$a_i(l) \le J \inf_{T\ge 0}\{C_{\mathrm{LR}} e^{vT - c_{\mathrm{LR}} l} + 2\tau(T)\}. \tag{13}$$

The scalar centering term is fixed by the conditional expectation. Split the remaining integral at T, apply (3) inside, and use the bound two outside. Inserting a certified tail, rather than a visually small time kernel, gives a physical input to Corollary 3.2. For example, $T = c_{\mathrm{LR}} l/(2v)$ when $v > 0$ and the stretched-exponential tail prove the premise of (11), after absorbing fixed polynomial factors into slightly smaller positive decay constants.

Proposition 4.1 (Normalized time truncation)**.** *Suppose $\tau = \tau(T) < 1$ and put $f_T = f\mathbf{1}_{[-T,T]}/(1-\tau)$, $\Phi_T = \Phi_{f_T}$, $M_i^T = \Phi_T(h_i) - e_i I$, and $k_i^T = |M_i^T|/(2J)$. Then Φ_T is unital completely positive, $0 \le k_i^T \le I$, and*

$$\|\Phi_T - \Phi\|_{\mathrm{cb}} \le 2\tau, \qquad \sum_i M_i^T = H - E_0 I, \tag{14}$$

$$\left\|M_i^T \Omega\right\| \le \frac{\tau}{1-\tau}\sigma_i, \qquad \sigma_i = \left\|(h_i - e_i I)\Omega\right\|. \tag{15}$$

In particular $H_T = \sum_i k_i^T \ge g(I-P)$. If

$$L_T = \frac{\tau}{1-\tau}\sum_i \frac{\sigma_i}{2J} < g, \tag{16}$$

then H_T has a unique ground state and gap at least $g - L_T$. This does not assert that its ground vector is Ω.

Proof. The total variation in density is exactly $\|f - f_T\|_1 = 2\tau$. Unitality, positivity and $\Phi_T(H) = H$ prove the first assertions. For every $|\omega| \ge \Delta$, the exact filter satisfies $\chi(\omega) = 0$, hence

$$|\chi_T(\omega)| = \left|\frac{-\int_{|t|>T} f(t)e^{it\omega}\,dt}{1-\tau}\right| \le \frac{\tau}{1-\tau}.$$

Apply this scalar bound only to the ground-to-excited column of h_i, where orthogonality of energy eigenvectors proves (15). The ground entry of M_i^T is zero. Since $\langle\Omega, |M_i^T|\Omega\rangle \le \left\|M_i^T\Omega\right\|$, the ground energy of H_T is at most L_T. The inequality $|M_i^T| \ge M_i^T$, summed, gives $H_T \ge g(I-P)$. The min–max principle puts its second eigenvalue at least g. These observations prove the gap and uniqueness assertion. □

The sum in (16) can grow with volume. One cannot use a fixed small τ to infer a volume-uniform version of this last conclusion. Also, $k_i^T\Omega$ is generally nonzero. Consequently neither $\sqrt{I - k_i^T}\,\Omega = \Omega$ nor the source's exact ground-preserving channel identity may be used for these time-truncated constraints.

There is a structural obstruction to repairing this by a universal finite-time exact filter. The Fourier transform of any integrable compactly supported time kernel extends to an entire function. If it vanishes on an open interval of forbidden energy differences, the identity theorem forces it to vanish everywhere, contradicting $\chi(0) = 1$. This does not rule out interpolation at a finite known spectrum; that is a different, spectrum-dependent problem.

5 Total truncation, linear state motion, and cut budgets

The source localizes all terms by a distance-adaptive construction, not by applying a fixed per-term error to an arbitrarily large sum. Let $S_0 \subseteq \Lambda$ have $|S_0| \le C_0 n^2$, $n \ge 2$, and set

$$r_i = \max\{r_0, \lfloor d_\Lambda(a_i, S_0)/2 \rfloor\}.$$

Terms in components disjoint from S_0 are kept exact. For all other terms let $\widetilde{k}_i = \mathsf{E}_{i,r_i} k_i$ and $\widetilde{H} = \sum_i \widetilde{k}_i$. The anchor multiplicity bound and the lattice ball bound imply, for constants depending only on the physical parameters and fixed p,

$$\begin{aligned} \left\| \widetilde{H} - H_F \right\| &\le \sum_i \left\| \widetilde{k}_i - k_i \right\| \\ &\le C|S_0| \sum_{d \ge 0} (d+1)^2 (1 + r(d)^\alpha) e^{-c r(d)^\alpha} \le C'|S_0| e^{-c' r_0^\alpha}, \end{aligned} \tag{17}$$

where $r(d) = \max\{r_0, \lfloor d/2 \rfloor\}$. For completeness, split the sum at $2r_0 + 2$. The first part is a polynomial in r_0 times $e^{-c r_0^\alpha}$. In the second, comparison to the integral and substitution $u = x^\alpha$ yield a polynomial times $e^{-c'' r_0^\alpha}$. Fixed polynomials are absorbed by reducing the positive decay constant. No factor $|\Lambda|$ is introduced. This general-α summation is already present in the source; its use here retains the sharper local modulus, rather than claiming a new shell-counting mechanism.

Theorem 5.1 (Linear ground-state error)**.** *Suppose $H_F \ge g(I - P)$, $H_F \Omega = 0$, $g > 0$, $\widetilde{H} \ge 0$, and $\left\| \widetilde{H} - H_F \right\| \le \epsilon < g/2$. Then $\widetilde{H}$ has a unique normalized ground vector $\widetilde{\Omega}$, energy $\widetilde{E} \in [0, \epsilon]$, and gap at least $g - 2\epsilon$. With its phase aligned,*

$$\delta = \tfrac{1}{2} \left\| P - |\widetilde{\Omega}\rangle\langle\widetilde{\Omega}| \right\|_1 = \left\| (I - P)\widetilde{\Omega} \right\| \le \frac{\epsilon}{g - \epsilon}, \tag{18}$$

$$\left\| \widetilde{\Omega} - \Omega \right\| \le \sqrt{2}\, \delta. \tag{19}$$

Proof. The ground trial Ω and positivity bound the lowest energy; Weyl's inequality bounds the second eigenvalue below by $g - \epsilon$. Let $Q = I - P$ and $V = \widetilde{H} - H_F$. Projecting the ground equation gives

$$(H_F - \widetilde{E}) Q \widetilde{\Omega} = -QV\widetilde{\Omega}.$$

On $Q\mathcal{H}$, the left operator has inverse norm at most $(g - \epsilon)^{-1}$. This proves the norm estimate. The pure-state trace-distance identity gives the equality for δ. Finally, phase alignment gives vector distance squared $2(1 - \sqrt{1 - \delta^2}) \le 2\delta^2$. □

This refines the source's valid energy-only square-root bound. It uses the full operator error, not just a small Rayleigh quotient. For a subsystem X of dimension $d_X = q^{|X|}$, the classical conditional-entropy continuity estimate [3], also proved in the source's preliminaries, gives

$$\begin{aligned} |S(X|Y)_\Omega - S(X|Y)_{\widetilde{\Omega}}| &\le C_{d_X}(\delta), \\ |I(X:Y)_\Omega - I(X:Y)_{\widetilde{\Omega}}| &\le 2C_{d_X}(\delta), \\ C_d(\delta) &= 2\delta \log d + (1+\delta)\log(1+\delta) - \delta \log \delta. \end{aligned} \tag{20}$$

At zero use the continuous value. Partial trace cannot increase δ. The spectator dimension never appears. Thus $\epsilon \le n^{-M}$, fixed $g > 0$, and $|X| \le Cn^2$ give $O(n^{2-M} + n^{-M} \log n)$ for this transfer alone. This is not a bound on the original entropy or mutual information itself.

Corollary 5.2 (Fixed-order logarithmic radii)**.** *For any fixed integer $p \geq 1$ and desired $M > 0$, a choice $r_0 = \lceil C(\log n)^{1/\alpha} \rceil$, $\alpha = p/(p+1)$, makes (17) at most $\min\{n^{-M}, g/4\}$, for sufficiently large fixed C. For a cut $X \subseteq S_0$ with b crossing edges, the source's cut budget for the localized terms satisfies*

$$B_X = 1 + \sum_{i:\,\text{crossing}} \log^2(e d_i^{\text{cut}}) \leq C'(1+b)(\log n)^{6/\alpha}. \tag{21}$$

Here d_i^{cut} is the smaller tensor dimension of the term's support across the cut, not the error $d_i(l)$.

Proof. Insert the chosen radius in (17) and absorb the factor n^2. A term crossing X cannot have its anchor ball disjoint from S_0, so every crossing radius in this construction is r_0. Its anchor is within distance r_0 of a crossing edge. There are at most $Cb(r_0+1)^2$ such anchors, including bounded multiplicity. Each ball contains at most $C(r_0+1)^2$ sites, so $\log(e d_i^{\text{cut}}) \leq C'(r_0+1)^2$. The product of the count and the squared logarithm is $C''b(r_0+1)^6$. This proves (21). □

The source's choice $p = 1$ gives exponent 12. The displayed alternative is $6 + 6/p$, for each fixed p, with possibly large p-dependent constants. It does not license setting $p = \infty$, varying p with system size without uniform estimates, or changing the source's final area-law exponent. Its amplification proof already uses a different, fixed high-order filter where needed.

6 Worked examples and necessary negative controls

Example 6.1 (An exact filtered one-spin decomposition)**.** On $\mathbb{C}^2$ take

$$H = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad h_1 = \begin{pmatrix} 1 & b \\ b & -1 \end{pmatrix}, \quad h_2 = \begin{pmatrix} -1 & -b \\ -b & 3 \end{pmatrix}, \qquad 0 \leq b \leq 1.$$

The safe common norm bound $J = 4$ follows from the absolute row sums. The ground vector is $(1,0)^t$, and $\Delta = 2$. Any filter of Proposition 2.1 with support in $[-1,1]$ kills the off-diagonal entries, giving

$$M_1 = \operatorname{diag}(0,-2), \quad M_2 = \operatorname{diag}(0,4), \quad H_F = \operatorname{diag}(0,3/4).$$

The actual gap $3/4$ exceeds the certified $g = 1/4$. This example checks the signed sum before absolute values and the normalization; it has no nontrivial entropy-area conclusion. For illustration only, the nonsmooth choice $h = \mathbf{1}_{[-a,a]}$ gives

$$\chi(\omega) = (1 - |\omega|/(2a))_+, \qquad f(t) = \frac{\sin^2(at)}{\pi a t^2}, \qquad \tau(T) \leq \frac{2}{\pi a T}.$$

It has the same positivity and exact spectral support, but not the Gevrey time tail needed in the source's locality amplification.

Example 6.2 (Scalar multiplier error is not operator error)**.** Take $H = \operatorname{diag}(0,1,4,10)$. Its six positive pairwise differences are distinct. There is a real even smooth compactly supported function m, with $\|m\|_\infty = 1$, such that $m(0) = 0$, $m(\pm 1) = -1$, and m is one at each of the other five pairs of differences: use disjoint smooth bumps about these twelve nonzero points. Put

$$C = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Its eigenvalues are $\sqrt{5}, -\sqrt{5}, 1, -1$, whereas $m(\operatorname{ad} H)(C)$ is the all-ones matrix minus the identity, of norm three. Thus

$$\|m(\operatorname{ad} H)\|_{\mathrm{op}\to\mathrm{op}} \geq 3/\sqrt{5} > 1.$$

The vanishing value at zero makes this a counterexample even for errors that preserve the normalization of a spectral filter at zero. It is not asserted to be a difference of two positive probability filters. In contrast, entrywise summation in an energy eigenbasis always gives $\|m(\operatorname{ad} H)(A)\|_{\mathrm{HS}} \leq \|m\|_\infty \|A\|_{\mathrm{HS}}$. The inverse Fourier transform of this smooth m is integrable; its L^1 norm necessarily accommodates the larger operator norm, in agreement with (12).

Example 6.3 (The channel square root cannot be removed)**.** Let $k = \operatorname{diag}(0, 1/2)$ and $l = \operatorname{diag}(t, 1/2)$, $0 < t < 1$. The channel in Corollary 3.2 acts on E_{12} with coefficient $1/\sqrt{2}$ for k and $(\sqrt{1-t} + \sqrt{t})/\sqrt{2}$ for l. Their difference divided by $\sqrt{t}$ tends to $1/\sqrt{2}$, while $\|k - l\| = t$. An operator-norm Lipschitz estimate uniform down to zero is therefore impossible, even for commuting constraints. The improvement in (11) removes only the preceding absolute-value square-root loss.

Example 6.4 (Conditional expectation loses annihilation)**.** For the two-qubit Bell vector $\Omega = (|00\rangle + |11\rangle)/\sqrt{2}$, the constraint $k = I - |\Omega\rangle\langle\Omega|$ kills Ω. Normalized expectation onto either one-qubit algebra is $\mathsf{E}(k) = 3I/4$, which does not. This exact computation explains why the source returns to its original constraints for amplification, after finishing the replica/scanner argument with localized ones.

Example 6.5 (Accumulated error cannot be replaced by a local one)**.** Let $H_F = \sum_{j=1}^N |1\rangle\langle 1|_j$. Replace each local projector by its real rotation through an angle θ, with $\epsilon = |\sin\theta| \in (0, 1)$. The local norm difference is ϵ, both global gaps equal one, but the squared ground-state overlap is $(1-\epsilon^2)^N$. The global difference has norm $N\epsilon$: its commuting single-site summands have eigenvalues $\pm\epsilon$. Consequently a fixed small per-term error does not control the global ground vector as $N \to \infty$. Equation (17) is an essential geometric input, not optional bookkeeping.

7 Where the global source proof still enters

The source is much more than a filtering construction. Its modules have the following roles; "imported" below means that this report does not supply an independent proof of the whole indicated step.

Initial tails and physical entropy. The preliminaries establish the norm and entropy conventions. The initial-tail module controls fluctuations of $-\log \rho_X$ on the positive spectrum of a marginal, centered at its entropy, with a crossing budget of the form B_X. Centering matters: a concentration estimate around an unknown entropy does not bound that entropy. The module's later buffered information argument is an imported input here.

Conditional powers and replicas. The conditional module uses positive spectral powers that vanish on the kernel, while inverse/imaginary powers use the separately specified identity extension on the kernel. These are different conventions. Its inequalities feed symmetric-replica branching and geometric-mean transports. The physical system, partitions, local dimensions, small parameters, and polynomial approximants are fixed before the replica number tends to infinity. Representation-theoretic polynomial factors can depend on these fixed dimensions. No finite Gram solve in this report makes such dependence uniform in volume.

Transport, comparators, scanner. The transport module differentiates a normalized metric-weighted norm and combines conditional-entropy gain with an energy budget. The comparator module uses the same auxiliary space and full-system metrics before and after comparison, rather than incompatible embeddings for individual blocks. The scanner imports these limits to obtain a sublinear collar information bound, for a region $|S_0| = O(n^2)$. Its terminal approximation error is transferred only after the replicas have disappeared. These are global analytic and uniformity obligations, not consequences of positivity in (7).

Exact-channel amplification. The full amplification module was read. It restarts with exact $G_i = \sqrt{I - k_i}$, $K_i = \sqrt{k_i}$, satisfying $G_i\Omega = \Omega$ and $K_i\Omega = 0$, and rate-one Poisson clocks for each label. Products place later events on the left. For retained versus omitted clocks its locality estimate controls an expectation of a norm on vectors of the form $B(\Omega \otimes \xi)$, not merely the norm of an expectation and not an unrestricted operator-norm difference between the two products. The excited square decays as e^{-gt}, using the exact gap and ground preservation. Typical-projector preparation may have large operator norm, but the source separately bounds its ground component and chooses the clock time to absorb the excited part.

Its fixed α is chosen very close to one to fit the sublinear collar exponent. The positive filter above permits such a fixed choice, but does not establish the scanner hypothesis. In particular one must not insert the time-truncated constraints of Proposition 4.1, or the spatially localized constraints of Example 6.4, into identities requiring exact annihilation. The source's polar-unitary step yields simultaneous decoupling from the whole distant algebra, with errors independent of its dimension. Pairwise decay or finitely many measured observables would not suffice.

Geometry and final entropy cancellation. The final module constructs two families of actual regions with ordered separation, repaired strips, and summable error budgets, including domains with holes. Given the required ordered mutual information estimates, the chain rule and purity yield $S(A) \le S(D) + \frac{1}{2}\sum_i \epsilon_i$, with $|D| \le C|\partial A|$. The report read this cancellation proof and the final assembly, but not the complete intervening geometric construction. These imported geometric hypotheses cannot be replaced by one favorable rectangular cut or by any of our finite examples.

No source correction is claimed by the local results. The original square-root estimates, normalization involving $\|f\|_1$, and $p = 1$ logarithmic budget are valid. They are refined here at selected interfaces, with their global roles and limit order retained.

8 Original Ram comparisons and literature status

Two original private text bodies were freshly read, locally and without transmitting them to public search. Their role is mathematical discipline for the physical norms, not a substitute area-law theorem.

The original Ram Calculus Fourier chapter [6] distinguishes weighted-contour absolute inversion from L^2 Plancherel, requires integrable majorants for interchange of convolution or moments, and warns that a variable symbol need not diagonalize an operator. It also separates nonvanishing multipliers from boundedly invertible ones. In our finite spin setting no contour deformation is needed: $f \in L^1(\mathbb{R})$ makes the physical time integral converge in operator norm, while Plancherel serves only to normalize the scalar probability kernel. Example 6.2 makes the distinction concrete even before an infinite-dimensional limit is considered. This is an application of those proof principles, not a claim that a Ram coordinate change proves complete positivity or the area law.

The original spectral-limits chapter [7] exhibits spectral pollution from compressed indefinite operators and separates compressed residuals from full physical residual norms. Its warning is directly relevant to Theorem 5.1: the required $\left\|\widetilde{H} - H_F\right\|$ is a full operator norm on the physical space. A residual tested on finitely many trial vectors cannot replace it. The theorem itself follows from the classical projected eigenvalue equation and not from a newly named spectral principle.

The primary public refresh on October 9, 2026 included [4, 5, 2, 3]. The abstract/version record of Anshu–Arad–Gosset concerns locally gapped frustration-free two-dimensional systems and a vertical bipartition. It does not certify the present source's arbitrary-cut, global-gap-only claim. Arad–Firanko–Jain, published February 5, 2026, uses a good approximate ground-space projector with an entanglement/contraction tradeoff; the abstract and opening definitions explicitly retain that input. A spectrally accurate finite matrix is not automatically such an AGSP. Kato's metadata establishes classical provenance for absolute-value continuity; the quantitative version used here has its own proof above. Winter's primary abstract/version record was checked, and the specific finite-dimensional continuity estimate was read in the source's proof. These access scopes do not amount to full audits of the external papers. The cached source is treated as a claimed proof, not as an accepted settlement inferred from its release or a bibliographic listing.

9 Scope, advantages, and remaining work

The public source architecture was mapped across all fifteen text files. Literal proof reading covered all of the quasilocal and amplification modules, the introduction, preliminaries, historical discussion and bibliography, plus specified substantial passages of the other modules. The read ledger records precise bounded display intervals and their unread complements. In particular, the full replica, transport, scanner, and geometric constructions were not independently audited. The two original private chapter text files were read in full; subordinate files included by the Fourier chapter and a figure included by the spectral chapter were not read. There was no whole-corpus review or formal-proof execution.

The independently executable checks exercise the rational filter fixture, noncommuting resolvent bounds, logarithmic-modulus examples, ground-state rotation, channel and normalized-trace counterexamples, and the four-level Schur multiplier. Exact algebraic cases and high-precision numerical diagnostics are counted separately in the owned receipt: 310 exact cases with 3,523 assertions, and 54 numerical cases with 108 assertions at 70 decimal digits. Numerical evaluations of logarithms are not interval certificates, and no finite fixture validates a uniform entropy, replica, or thermodynamic assertion. The local theorems rest on their proofs and stated physical premises.

Benefits and costs. The positive kernel fixes its L^1 normalization at one, and the logarithmic absolute-value modulus avoids an unnecessary first square-root loss. The linear state bound then converts a genuinely summed error into a sharper subsystem-continuity budget. These are notable improvements at an actual source step, rather than a generic matrix representation. Their costs remain real: the ground expectations e_i and a valid global gap are required; time and spatial tails need physical constants; large fixed Gevrey order can have poor constants; and final root operations have an unavoidable endpoint loss. No runtime or overall area-law constant improvement is measured.

Extensions with explicit missing premises. A spatially heterogeneous construction can minimize radii subject to $\sum_i d_i(r_i) \le \epsilon$, provided actual anchor geometry and each tail bound are supplied. Certified quadrature could be inserted through an L^1 kernel or a directly certified

completely bounded error, but scalar interpolation alone is insufficient. Higher-dimensional bounded-growth graphs change the shell and cut counts; they do not automatically inherit the source's two-dimensional geometry or entropy cancellation. A degenerate ground space would need a different centering/annihilation construction, because Ph_iP need not be scalar. Extending the present formulas without that modification would be incorrect.

Contribution boundary. The C3 assessment concerns the combined loss-reduced physical filtering, channel, and summed-error interface, not historical priority for autocorrelation, Kato-type continuity, or perturbation theory. Applicability remains L3. No improved global entropy theorem, independent verification of the source's entire proof, or full many-body numerical certificate is asserted. The useful next step is to quantify the actual filter and propagation constants for a fixed model and fixed Gevrey order, then compare complete feasible budgets rather than exponents alone.

References

[1] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026, cached release manuscript. Public source. The byte-identity provenance is separate from proof validity.

[2] T. Kato, *Continuity of the map $S \mapsto |S|$ for linear operators*, Proc. Japan Acad. 49 (1973), 157–160. DOI. Primary metadata checked; local modulus proved in this report.

[3] A. Winter, *Tight uniform continuity bounds for quantum entropies: conditional entropy, relative entropy distance and energy constraints*, Commun. Math. Phys. 347 (2016), 291–313. arXiv:1507.07775v6.

[4] A. Anshu, I. Arad and D. Gosset, *An area law for 2D frustration-free spin systems*, STOC 2022, 12–18. arXiv:2103.02492v3. Abstract/version record and opening theorem scope checked; no full proof audit.

[5] I. Arad, R. Firanko and R. Jain, *Area Laws and Tensor Networks for Maximally Mixed Ground States*, Commun. Math. Phys. 407 (2026), article 54. DOI. Abstract and opening AGSP assumptions checked; proof not independently audited.

[6] M. Subbarao, *Ram Calculus*, volume 1, revision 2, original chapter file `ch07-ramcs-rev2.tex`. Confidential local source, literal body lines 1–373 read; subordinate included files not read.

[7] M. Subbarao, *Ram Calculus*, volume 21, original chapter file `ch01f-spectral-limits.tex`. Confidential local source, literal body lines 1–242 read; included figure not inspected.